Notes - PreBoot

Sunday, June 28, 2026 11:30 AM

Approaches
- SfM (Surface from Motion)
  - Focused on creating a 3D model from 2D input
- SLAM (Similtaneous Localization and Mapping)
  - Detect the position of the image provider and where it is going. Mapping is required to do this. SfM can be part of this pipeline.
- VPR (Visual Place Recognition)
  - This does not create a 3D model. It is exclusively focused on matching placing between snapshots.
  - This can be optimized by storing images as "compact visual representations". I assume this means vectors, spare maps, or interest/descriptor pairs.

    VPR is often a component in slam. It is used for LOOP closure.

    Super focused on whole image matching within a candidate structure. Global descriptors rather than the locals used by superpoint/superglue for pairing.
    VPR is the course candidate finding layer in SfM.

Trends
- DUSt3R/MASt3R
  - Utilization of 3D space understanding models as priors to SLAM approach
  - Dig into this with more detail: https://chatgpt.com/g/g-p-6a3989f5909c8191b48522794c68aea3-umbc/c/6a413d55-b678-83ea-a5f4-8f5eecfbc23b

    Part of a larger trend of incorporating 3D priors (<- figure out why that is important, still a little bit out of reach atm)
    Dynamic Visual SLAM using a General 3D Prior
- 4D SLAM
  - The generated map is itself dynamic. It can recognize static objects such as light poles, but also dynamic objects such as cars and pedestrians.
  - First seen here:
    - RU4D-SLAM: Reweighting Uncertainty in Gaussian Splatting SLAM for 4D Scene Reconstruction
    - DROID-SLAM in the Wild

      Both of these are novel extension in the field though. To learn more dig into more foundational papers.

    Key point: Dynamics in sensor capture are not just noise to remove, they can also be import data model and capture.

Challenges
- Scale-Drift
  - Scale is hard to infer from a single camera because it lacks depth. A close up of a small object and a far shot of a large object look similar.
  - This ambiguity causes drift over time. Scale is determined on image at a time.

    Solution: SCE-SLAM: Scale-Consistent Monocular SLAM via Scene Coordinate Embeddings
    - Scale is set canonically across the entire scene and forced to be consistent across frames.
    - This makes the bundle adjustments a learned property of the network? It adds a global constraint to the usually imposed global constraints.

      geometry-modulated attention?
- Input degradation
  - Rather than testing on clean input what about motion blur and lack of camera focus?
  - Solution
- Priors
  - NeRF w/ no pose information
    - https://openaccess.thecvf.com/content/CVPR2023/papers/Bian_NoPe-NeRF_Optimising_Neural_Radiance_Field_With_No_Pose_Prior_CVPR_2023_paper.pdf
- Speed
  - COLMAP
    - The current standard for high accuracy in SfM is COLMAP. However this does not scale well to large scenes of thousands of images.
    - FASTMAP: Revisiting Dense and Scalable Structure from Motion "
      - "In such a scenario, which is our focus, the current state-of-the-art is still held by COLMAP [45]."
      - "Learning-based methods such as DUSt3R [53] rely on COLMAP to provide pseudo ground truth for training."
        - ^Can duster be improved by swapping out the base method?
  - GOLMAP
    - Improvement on COLMAP "through global (instead of incremental) SfM, but still takes many hours to converge on large scenes." (FASTMAP)
  - FASTMAP
    - 1-2 order of magnitude faster but is GPU based? Keep similar accuracy levels?
- Mobile Readyness
  - AR is a field that could contain inspiration for here.
  - VIO - Visual Inertia Tracking
    - Of interest in 2025/2026 for VR applications. Explicitly focused on combining phone IMU with visual feature tracking. It IS an odometry method.
    - Feels odometry esk?

Key Ideas
- Sparsity
  - https://www.youtube.com/watch?v=SbU1pahbbkc

    The full dataset is not needed to make inferences. Just key points. How much is enough? And how can this be optimized to improve speed/reliability.
- Loop Closure
  - This is related to pose estimation in construction of the map. Incorrect detection of visiting an already visited place can lead to drift and creation of novel pathways instead of improvements on existing structures.

    Fastest: BoVW, NetVLAD
    Mid Fast: VPR
    Mid Slow: SIFT, ORB, ETC
    Slowest: Dense Mapping Stuff

  - CHATGPT
    - Distinct Stages
      - Loop Detection
      - Loop Verification
      - Loop Correction
    - Challenges
      - Viewpoint changes
      - Illumination changes
      - Dynamic objects
      - Seasonal / long-term changes
      - Perceptual aliasing
      - Scalability
    - Historic Approaches
      - Local features - intersest points + descriptors
      - Bag of words
      - Geometric verification
    - Current Approaches
      - Global descriptors - each image/scene is condensed down to a singular vector.
        - NetVLAD & Faiss
          1. i) Convert current observation into a descriptor vector.
          2. ii) Search stored descriptors for nearest neighbors.
          3. iii) Treat the nearest prior places as loop candidates.
          4. iv) Verify the candidates before accepting them.
      - VPR
        - If you've been here before you've been hear before.

Misc
- Neural Radiance Fields
  - Extremely efficient compression of spaces
  - 15GB mesh -> 5mb radiance field
    - Less space than the composing images

    Can be thought as a function that returns voxels based on input ray
- Neural Surfel Reconstruction: Addressing Loop Closure Challenges in Large-Scale 3D Neural Scene Mapping
  - Core contribution: making neural surfels compatible with loop closure in SLAM

    Corrects drift my optimizing the vectors contained in neural surfels. This is an alternative to optimizing a singular neural radiance field.

    Claims a 90% reduction in model file size and 20% reduction in reconstruction error compared to "classical methods." I am unfamiliar with neural representation contruction methods so these comparisons could not be reviews for meaningful review of accuracy on that claim.

    Note: not suitable for real-time.
- Sources of changes
  - Illumination
  - Weather
  - Lighting factors
  - Environmental differences
- TSDF
  - Truncated Signed Distance Function
  - Math used to create depth informed voxel maps

    Pros:
    - Improved depth measurements
    - Continuous surfaces
    - Incremental reconstruction (overlayed/averaged depth frames)

      Powerful for scene reconstruction (depth)
      Better scenes = better localization

VPR is built for image mapping.

Siamese Networks
- Input: two images
- Output: are they the same place?
- 2025 nature paper on this approach (Link)

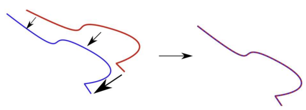
Cross-attention in Transformers

Semantic Approaches
- Useful in sparse characteristic areas. A comparison can be done on objects as the words or descriptors

OverlapNet - A NN dedicated specifically to overlap detection in lidar based SLAM systems. (2020)

Approaches: (These came from a chart in OverlapNet)
- Bag of words
- NN
- Histogram?
- SuMa
- M2DP

Approaches from Think Autonomous - Introduction to Loop Closure Detection in SLAM:
- Closure Detection - Are we in a loop? Bring it together.
  - Iterative Closest Point (ICP)
    - This focuses on finding corresponding points between point clouds and then estimating a transformation to align them.
    - Least squares approach for cost estimates.

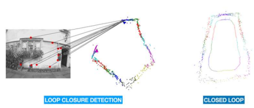


The Iterative Closest Point (wikipedia)

- Closure Candidates - We've been here before. Don't create a new path.
  - Bag-of-Words
    - Represent each image as words
  - FAB-MAP.
    - Use Bag-of-Words to estimate likelyhood of previous visit. (Bayesian)

Feature Matching and how it can introduce errors (source)

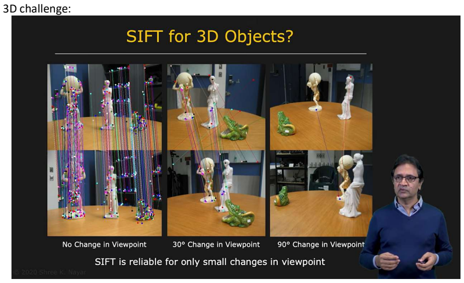
Map Correction



Loop Closure Detection and Map Correction (source)

SLAM has different methods for this but it is a core part.

Interest Points / Descriptors
- Uses:
  - Picture matching across changing orientations, scales, lighting conditions
    - -> matching despite occlusions
    - -> stitching images together based on matching interest points
    - -> construction of collages from multiple images (next step past stitching)
  - How does this transform from 2D to 3D?
    - The 3D challenge:



(Pay attention to the difference in magnitude of matches.)

3D objects look radically different based on viewing angle. This is more than a brightness/gradiated scaling distortion.